

$$f(x) = -e^x + \sqrt{1-x} \quad | \quad x \geq 1$$

$$1-x \geq 0$$

Question 1: The domain of the function $f(x) = -e^x + \sqrt{1-x}$ equals

- ☒ A) $(-\infty, 1)$ ☐ B) $(-\infty, 1]$ ☐ C) $(1, \infty)$ ☐ D) $[1, \infty)$ ☐ E) None

Question 2: The domain of the function $f(x) = \frac{\sin x}{\sin x}$ equals $\sin x = 0$
 $x = 0, \pi, 2\pi$

- ☒ A) $\mathbb{R}$ ☐ B) $\mathbb{R} - \left\{ \frac{2n+1}{2}\pi, n \in \mathbb{Z} \right\}$ ☐ C) $\mathbb{R} - \{n\pi, n \in \mathbb{Z}\}$
☐ D) $\mathbb{R} - \{2\pi/3\}$ ☒ E) None

Question 3: The set of real solutions of the equation $e^{2x} + 5e^x + 6 = 0$ equals

- ☒ A) $\emptyset$ ☐ B) $\{2, 3\}$ ☐ C) $\{-2, -3\}$ ☐ D) $\{\ln(2), \ln(3)\}$ ☐ E) None

Question 4: For the functions $f(x) = 1 + e^{|x|}$ and $g(x) = \sin(x^2)$ which of the following is true

- ☐ A) f is odd and g is even ☐ B) Both functions are odd
☐ C) f is even and g is odd ☒ D) Both functions are even ☐ E) None

$$2 \ln(x) = (\ln x + \ln x)$$

Question 5: The domain of the function $f(x) = 2 \ln(x) - \ln(x^2)$ equals

- ☐ A) $[0, \infty)$ ☒ B) $(0, \infty)$ ☐ C) $\mathbb{R}$ ☒ D) $[1, \infty)$ ☐ E) None

$$f(x) = 2 \ln(x) - (2 \ln x)$$

Question 6: How many vertical asymptotes does the function $f(x) = \frac{x}{x-x^3}$ possess?

- ☐ A) 2 ☐ B) 1 ☐ C) 0 ☒ D) 3 ☐ E) None of the stated answers

$$f(x) = 1 + e^{|x|}$$

$$f(-x) = 1 + e^{|-x|}$$

$$= 1 + e^{|x|}$$

$$f(x) = \sin(x^2)$$

$$f(-x) = \sin((-x)^2)$$

$$= \sin(x^2)$$

even

Question 7: For each $x \in (-1, 1)$, we have $\cot(\sin^{-1}(x)) =$

- A) $\sqrt{1-x^2}$ B) x C) $\frac{\sqrt{1-x^2}}{x}$ D) $\frac{x}{\sqrt{1-x^2}}$ E) None

Question 8: The set of real solutions of the equation $2^x = 8^x$ equals

- A) $\{2\}$ B) $\emptyset$ C) $\{-2\}$ D) $\{0\}$ E) None

Question 9: The range of the function $f(x) = (x-1)^2 + 2$ equals

- A) $[-2, \infty)$ B) $(-2, \infty)$ C) $[2, \infty)$ D) $(2, \infty)$ E) None

Question 10: Which of the following real functions is NOT one-to-one

- A) $f(x) = \sin(x)$ B) $f(x) = \ln(x)$ C) $f(x) = e^x$
D) $f(x) = x^5$ E) None

Question 11: $\lim_{x \rightarrow 0} \left(\sin \left(\frac{\sqrt{x^3+1} - 1}{x} \right) \right)$ equals

- A) d.n.e B) 0 C) 2 D) ∞ E) None

Question 12: $\lim_{x \rightarrow -5+} \left(\frac{x^2}{x+5} \right)$ equals

- A) 0 B) 1 C) $-\infty$ D) $+\infty$ E) None

$$\begin{array}{l|l} \lim_{x \rightarrow +5} \frac{25}{10} = 2.5 & \lim_{x \rightarrow -5+} \left(\frac{x^2}{x+5} \right) \\ \hline \lim_{x \rightarrow -5} \frac{25}{0} = \infty & \lim_{x \rightarrow +5} \left(\frac{25}{10} \right) = 2.5 \\ & \lim_{x \rightarrow -5} \frac{25}{0} = \infty \end{array}$$

Question 13: For the functions $f(x) = \log_2 x$ and $g(x) = e^x + 1$, derive $(f \circ g)(x)$ and $(f \circ g)(0)$

$$(f \circ g)(x) = f(g(x)) = f(e^x + 1) = \log_2(e^x + 1)$$

$$f(g(0)) = g(0) = e^0 + 1 = 1 + 1 = 2$$

$$f(g(0)) = f(2) = \log_2(e^2 + 1)$$

$$\cancel{\log_2 e^2} \rightarrow \cancel{\log_2 1}$$

Question 14: For the real function $\underline{\underline{y = e^{e^x}}}$, find the inverse function

$$f(x) = e^{e^x}$$

$$f(x) = e^{e^x} = y$$

$$\ln(e^{e^x}) = \ln y$$

$$e^x = \ln y$$

$$x = \ln(\ln y)$$

Question 13: For the functions $f(x) = \log_2 x$ and $g(x) = e^x + 1$, derive $(f \circ g)(x)$ and $(f \circ g)(0)$

$$(f \circ g)(x) = f(g(x)) = f(e^x + 1) \quad (1M)$$

$$= \log_2(e^x + 1) \quad (1M)$$

$$(f \circ g)(0) = \log_2(2) = 1 \quad (1M)$$

Question 14: For the real function $y = e^{e^x}$, find the inverse function

$$\ln y = e^x \ln e$$

$$\ln y = e^x \quad (1M)$$

$$\ln(\ln y) = x \ln e$$

$$\Rightarrow x = \ln(\ln y) \quad (1M)$$

$$\Rightarrow f^{-1}(x) = \ln(\ln x) \quad (1M)$$

